

Theorem 1.1: a formalization blueprint

Sparse-incidence width comparison

Lean 4.31.0

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1 Scope

The main path follows Theorem 1.1 of `main.tex`. Lean proves the final comparison by a different recursion, recorded in Section 8. Checkmarks identify checked Lean declarations. The article’s algorithmic and complexity results are outside this blueprint.

2 Finite hypergraphs and structural parameters

Definition 2.1 (Hypergraphs). A manuscript hypergraph is finite, has nonempty edges, and has no isolated vertices. Its edges form a set of vertex sets. The input convention takes H to be nonempty and connected, while restrictions may be empty or disconnected. The primal graph joins distinct vertices in a common edge, and the incidence graph joins a vertex to an edge containing it. For $W \subseteq V(H)$, $H[W]$ intersects each edge with W , discards empty intersections, and merges equal intersections. Lean instead retains distinct edge labels that describe the same vertex set, including after restriction.

In Lean the finite label sets are subsets of the natural numbers. Values of the edge map on inactive labels are immaterial. In particular, transferring from $H[V(H)]$ to H uses agreement on active edges, not literal equality of the entire edge-map structures.

Definition 2.2 (Incidence degeneracy and primal independence). Let $\mathcal{C}(H)$ consist of the nonempty primal connected components, let $d(H)$ be the degeneracy of the incidence graph, and let $\alpha(H)$ be the maximum size of a primal independent set. In the manuscript, $d(H)$ is denoted $\mu(H)$. Lean's auxiliary structural factor is $A(H) = 1 + \min\{d(H), \ln(2 + \alpha(H))\}$.

Definition 2.3 (The manuscript parameter).

$$\lambda(H) = \min\{\mu(H), \max\{1, \log_2 \alpha(H)\}\}.$$

All unqualified logarithms in the manuscript are binary. Lean also uses

$$L(H) = 1 + \max_{K \in \mathcal{C}(H)} \min\{d(H[K]), \ln(2 + \alpha(H[K]))\},$$

with maximum zero when H is empty. For nonempty H it proves $L(H) \leq A(H) \leq 4\lambda(H)$. The componentwise factor is internal to the Lean proof and is not the manuscript's λ .

Lemma 2.4 (Structural heredity). *For every restriction, $d(H[W]) \leq d(H)$ and $\alpha(H[W]) \leq \alpha(H)$.*

Proof. Embed the restricted incidence graph into the original graph and use the induced-subgraph characterization of degeneracy. A primal independent set of a restriction remains independent in the original hypergraph. Thus $\lambda(H[W]) \leq \lambda(H)$ whenever the parameters are defined. $\square$

3 Covers, tree decompositions, and widths

Definition 3.1 (Fractional edge-cover cost). For a demand x , a cover is a nonnegative edge vector y with $\sum_{e \ni v} y_e \geq x(v)$ for every vertex. Its optimum cost is

$$\rho_H^*(x) = \inf_y \sum_e y_e, \quad \rho_H^*(X) = \rho_H^*(\mathbf{1}_X).$$

An edge-dominated modular weighting is a nonnegative vertex vector w with $\sum_{v \in e} w_v \leq 1$ on each hyperedge; write $b_w(X) = \sum_{v \in X} w_v$.

Lemma 3.2 (Cover duality). *For a unit demand $x : V(H) \rightarrow [0, 1]$ there is an edge-dominated modular w such that*

$$\rho_H^*(x) = \sum_v w_v x(v), \quad \sum_v w'_v x(v) \leq \rho_H^*(x) \quad \text{for every admissible } w'.$$

Proof. Construct a supporting linear functional for the sublinear cover-cost functional. Its values on singleton vectors yield w . Feasible covers give the reverse inequality by exchanging the two finite sums. $\square$

Lemma 3.3 (Restriction preserves cover cost). *If $X \subseteq W$, then $\rho_{H[W]}^*(X) = \rho_H^*(X)$.*

Proof. In the manuscript, aggregate weights on edges with equal nonempty intersections with W , and lift a restricted cover by choosing one original edge for each restricted edge. For Lean's labelled edges, retain surviving labels and extend the other weights by zero. $\square$

Definition 3.4 (Tree decompositions and width parameters). A tree decomposition is a finite nonempty tree with vertex-subset bags, covering every hyperedge and vertex, such that the bags containing any fixed vertex form a connected subtree. For a bag cost b , set

$$\text{opt}_H(b) = \inf_D \max_{t \in D} b(B_t).$$

Then $\text{fhw}(H) = \text{opt}_H(\rho_H^*)$ and $\text{adw}(H) = \sup_w \text{opt}_H(b_w)$ over admissible modular weights. For $\text{subw}(H)$ take the supremum instead over normalized, monotone, submodular, edge-dominated set functions.

Lemma 3.5 (Width hierarchy and heredity). *One has $\text{adw}(H) \leq \text{subw}(H) \leq \text{fhw}(H)$, and $\text{adw}(H[W]) \leq \text{adw}(H)$ and $\text{fhw}(H[W]) \leq \text{fhw}(H)$.*

Proof. Modular functions are submodular. Supporting modular weights bound each submodular bag value by its fractional cover cost. Restricting decompositions and extending admissible weights gives heredity. $\square$

Lemma 3.6 (Component reduction). *Each of $\text{fhw}(H)$ and $\text{adw}(H)$ is the maximum of its values on $H[K]$ for $K \in \mathcal{C}(H)$, with value zero in the empty case.*

Proof. Join the disjoint component trees, retaining their bags and internal paths. A hyperedge lies in a single component. Heredity gives the reverse bounds; passing to infima avoids requiring optimal decompositions to be attained. $\square$

4 Balanced separators

Definition 4.1 (Fractional and integral balance). Let $\gamma : E(H) \rightarrow [0, 1]$ have positive total mass $M = \sum_e \gamma_e$. For a unit demand x , the distance between vertices is the minimum of 1 and the x -length of a simple primal path, counting both endpoints; absent paths have distance 1. The distance $d_x(e, f)$ is the minimum over endpoint choices in e and f . The vector x is fractionally (γ, θ) -balanced if

$$\sum_f \gamma_f d_x(e, f) \geq (1 - \theta)M \quad (e \in E(H)).$$

A set $S \subseteq V(H)$ is integrally balanced if every component K of $H[V(H) \setminus S]$ satisfies

$$\sum_{e: e \cap K \neq \emptyset} \gamma_e \leq \theta M.$$

Definition 4.2 (Hereditary separator costs). Let $\beta_H^\theta(\gamma)$ be the infimum of $\rho_H^*(x)$ over fractional balanced separators. Define $\text{fsep}_\theta(H)$ by taking the supremum over all $H[W]$ and admissible positive edge weights. Define $\text{sep}_\theta(H)$ analogously using integral separators and their fractional cover cost. Both hereditary suprema explicitly include zero.

Lemma 4.3 (Convex fractional separators). *For a nonzero weighting γ and $0 < \theta < 1$, the set of fractional (γ, θ) -balanced separators is nonempty, compact, and convex.*

Proof. The truncated edge distance is the minimum of 1 and finitely many path costs. Each path cost is $\sum_{v \in V(Q)} x(v)$, so the distance is continuous and concave in x . Each balance constraint is a closed convex superlevel set in $[0, 1]^{V(H)}$. The all-one vector satisfies every constraint. Lean proves these properties inside its minimax argument but has no standalone declaration with this statement. $\square$

Lemma 4.4 (Indicator equivalence). *For $S \subseteq V(H)$, admissible γ , and $0 < \theta < 1$, S is integrally (γ, θ) -balanced exactly when $\mathbf{1}_S$ is fractionally (γ, θ) -balanced.*

Proof. Indicator distances detect connectivity after deleting S . For an edge meeting a surviving component, its weighted distance sum is total mass minus the mass of edges meeting that component. Edges wholly deleted have distance one to every edge. $\square$

Lemma 4.5 (Balanced bag). *Every tree decomposition and every admissible γ have a bag that is an integral $(\gamma, 1/2)$ -balanced separator.*

Proof. Assign every edge's mass to one covering bag and choose a weighted median of the tree. A residual primal component can meet bags in only one branch of the tree after deleting the median. $\square$

Lemma 4.6 (Recursive separator decomposition). *Suppose $0 < \eta < 1$ and $r \geq 0$. If every restriction $H[W]$ and every nonzero edge weighting on $H[W]$ admit an integral (γ, η) -balanced separator S with $\rho_{H[W]}^*(S) \leq r$, then*

$$\text{fhw}(H) \leq \frac{2(2 - \eta)}{1 - \eta} r.$$

Proof. Put $\Lambda = 2r/(1 - \eta)$ and induct on $|W \setminus Z|$ for $Z \subseteq W \subseteq V(H)$ with $\rho_H^*(Z) \leq \Lambda$. Construct a decomposition of $H[W]$ containing Z in one bag and with every bag of H -cover cost at most $\Lambda + 2r$. If $W = Z$, one bag suffices.

Otherwise, if $Z \neq \emptyset$, cover Z optimally and use the resulting edge weights γ to choose a separator S of cost at most r . If $Z = \emptyset$, take $S = \emptyset$. Choose for each $v \in W \setminus Z$ an incident edge e_v and put

$$\pi(e) = \frac{|\{v \in W \setminus Z : e_v = e\}|}{|W \setminus Z|}.$$

The weighting π has total mass 1. Choose a (π, η) -balanced separator R of cost at most r and set $B = Z \cup S \cup R$. Then $\rho_H^*(B) \leq \Lambda + 2r$.

For each component K of $H[W] - B$ set $Z_K = B \cap N_{P(H[W])}(K)$ and $W_K = K \cup Z_K$. If $Z = \emptyset$, $Z_K \subseteq R$, so $\rho_H^*(Z_K) \leq r \leq \Lambda$. Otherwise let C be the component of $H[W] - S$ containing K . The inclusion $Z_K \subseteq S \cup R \cup (Z \cap C)$ and the balance of S give

$$\rho_H^*(Z_K) \leq 2r + \eta \rho_H^*(Z) \leq 2r + \eta \Lambda = \Lambda.$$

If D is the component of $H[W] - R$ containing K , the balance of R gives $|K|/|W \setminus Z| \leq \sum_{e \in D \neq \emptyset} \pi(e) \leq \eta < 1$. Hence $|W_K \setminus Z_K| = |K| < |W \setminus Z|$. Apply induction to each (W_K, Z_K) and attach its designated bag to the new bag B . Running intersection follows from $B \cap W_K = Z_K$. An edge not contained in B lies in one child because it is a primal clique. Finally apply the invariant with $(W, Z) = (V(H), \emptyset)$ to obtain $\Lambda + 2r = 2(2 - \eta)r/(1 - \eta)$. $\square$

5 Adaptive width controls fractional separators

Lemma 5.1 (Fractional-cover minimax). *For admissible γ and $0 < \theta < 1$, there is an admissible modular weight w such that every fractional (γ, θ) -balanced x obeys*

$$\beta_H^\theta(\gamma) \leq \sum_v w_v x(v).$$

Proof. Express path distances as finite minima over path supports, with an extra constant-one option. The feasible separator set is compact and convex by Lemma 4.3. The modular-weight cube is also compact and convex. Apply Sion’s minimax theorem to the bilinear pairing and use cover duality to identify the optimum. $\square$

Lemma 5.2 (Hereditary fractional separator bound). *In particular, for $\theta = 1/2$,*

$$\text{fsep}_{1/2}(H) \leq \text{adw}(H).$$

The Lean declaration proves this for every $1/2 \leq \theta < 1$.

Proof. Use the minimax modular weight and a balanced bag in any decomposition to bound the local fractional optimum. Pass to the decomposition infimum, then use adaptive-width heredity and take the hereditary supremum. $\square$

6 Rounding and its consequences

Theorem 6.1 (The proved rounding input). *Let $0 < \varphi < 1$, let $0 \leq \gamma_e \leq 1$, and let x be fractionally (γ, φ) -balanced. Put $c = \rho_H^*(x)$, $T = \{v \in V(H) : x(v) \neq 0\}$, and $\eta = (1 + 3\varphi)/(2 + 2\varphi)$. Assume the empty set is not already integrally (γ, η) -balanced. Then there is $S \subseteq T$ that is integrally (γ, η) -balanced with*

$$\rho_H^*(S) \leq (\min\{8 + 4 \ln \alpha(H[T]), 6\mu(H)\} + 1) \frac{44 + 8 \log_2(c/(1 - \varphi))}{1 - \varphi} c.$$

Proof. The Lean proof establishes the existence conclusion of Theorem 5.6 in Korchemna–Lokshtanov–Saurabh–Surianarayanan–Xue, [arXiv:2409.20172v1](https://arxiv.org/abs/2409.20172v1). The nontrivial-branch hypothesis is an additional restriction of our use of the cited result. It avoids the printed expression’s negative-logarithm issue at small positive costs. The proof includes both bounds in their two-terminal Theorem 4.1. It repairs the radius budget in Lemma 5.9 while preserving the constants. The cited running-time guarantee is not formalized here. $\square$

Proposition 6.2 (The manuscript’s rounding bound). *There is a universal $C_0 > 0$ such that for every J , nonzero weighting $\gamma : E(J) \rightarrow [0, 1]$, and fractional $(\gamma, 1/2)$ -balanced x , there is an integral $(\gamma, 5/6)$ -balanced $S \subseteq V(J)$ with*

$$\rho_J^*(S) \leq C_0 \lambda(J) \rho_J^*(x) \log_2 \max\{2, \rho_J^*(x)\}.$$

Proof. If $\emptyset$ is already balanced, take $S = \emptyset$. Otherwise $q = \rho_J^*(x) > 0$, since $q = 0$ forces $x = 0$ and Lemma 4.4 makes $\emptyset$ balanced. Specialize Theorem 6.1 to $\varphi = 1/2$. Its structural factor is at most $7\mu(J)$ and at most $9 + 4 \ln \alpha(J)$, hence at most a constant times $\lambda(J)$. Also $88 + 16 \log_2(2q) = O(\log_2 \max\{2, q\})$ on this nontrivial branch. This sharper specialization is the manuscript’s argument. Lean proves the rounding input and a regularized consequence, but has no separate declaration for this displayed bound. $\square$

7 The main comparison

Theorem 7.1 (Theorem 1.1: main width comparison). *There is a universal constant $C > 0$ such that every input hypergraph H with $\text{adw}(H) \geq 2$ satisfies*

$$\text{fhw}(H) \leq C\lambda(H)\text{adw}(H)\log_2 \text{adw}(H).$$

When $\text{adw}(H) < 2$, $\text{fhw}(H) \leq C\lambda(H)$ with the same constant. The Lean statement assumes nonemptiness and does not require connectivity.

Proof. This is the manuscript's proof. Put $a = \max\{2, \text{adw}(H)\}$. For any $J = H[W]$ and nonzero γ , compactness and continuity give a minimizing fractional half-balanced separator x . Lemma 5.2 gives

$$\rho_J^*(x) = \beta_{J,1/2}(\gamma) \leq \text{fsep}_{1/2}(H) \leq \text{adw}(H) \leq a.$$

Structural heredity gives $\lambda(J) \leq \lambda(H)$. Proposition 6.2 supplies an integral 5/6-balanced separator of cost at most $r = C_0\lambda(H)a\log_2 a$. This follows from monotonicity of $q\log_2 \max\{2, q\}$ for $q \geq 0$. The recursive lemma with $\eta = 5/6$ yields

$$\text{fhw}(H) \leq 14r = 14C_0\lambda(H)a\log_2 a.$$

For $\text{adw}(H) \geq 2$ this is the first case. For $\text{adw}(H) < 2$ it is a constant times $\lambda(H)$. Lean proves the theorem statement through the alternative route in Section 8. This proof has no separate Lean completion marker. $\square$

8 The Lean route to the same comparison

The checked Lean argument uses a regularized rounding estimate and a separator-plus-one-vertex recursion. Its intermediate inequality differs from Lemma 4.6. The manuscript's two-separator induction and the sharper specialization in Proposition 6.2 have no one-to-one Lean declarations.

Lemma 8.1 (Regularized balanced-separator rounding). *There is a universal $C_0 > 0$ such that a fractional $(\gamma, 1/2)$ -balanced x admits an integral $(\gamma, 5/6)$ -balanced $S \subseteq \text{supp } x$ with*

$$\rho_H^*(S) \leq C_0(1 + \min\{d(H), \ln(2 + \alpha(H[\text{supp } x]))\})(\rho_H^*(x) + 1)\ln(2 + \rho_H^*(x)).$$

Proof. Use the empty separator when it is balanced. In the other case specialize the proved rounding input at $\varphi = 1/2$ and bound its factors. Lean chooses $C_0 = 9(120/\ln 2)$. $\square$

Lemma 8.2 (Lean's rooted recursion). *Fix $0 < \eta < 1$ and $r > \text{sep}_\eta(H)$, and put $\Lambda = (r+1)/(1-\eta)$. For $Z \subseteq W \subseteq V(H)$ with $\rho_H^*(Z) \leq \Lambda$, there is a decomposition of $H[W]$ with one bag containing Z and all bags of H -cover cost at most $\Lambda + r + 1$.*

Proof. Induct on $|W \setminus Z|$. If $W = Z$, take one bag. Otherwise choose $u \in W \setminus Z$ and an integral separator S of cost below r from an optimal cover of Z , taking $S = \emptyset$ if Z is empty. Set $B = Z \cup S \cup \{u\}$. For each component K of $H[W] - B$, set $Z_K = B \cap N_{P(H[W])}(K)$ and $W_K = K \cup Z_K$. Balance gives $\rho_H^*(Z_K) \leq r + 1 + \eta\Lambda = \Lambda$. The pivot makes $|K| < |W \setminus Z|$, so induction applies to each child. Join child roots to the bag B . $\square$

Lemma 8.3 (Lean's recursive separator bound). *For $0 < \eta < 1$,*

$$\text{fhw}(H) \leq \frac{2-\eta}{1-\eta}(\text{sep}_\eta(H) + 1).$$

Proof. Apply the rooted invariant to $W = V(H)$ and $Z = \emptyset$, then let r decrease to $\text{sep}_\eta(H)$. $\square$

Lemma 8.4 (Hereditary rounding in Lean). *There is a universal $C_1 > 0$ such that every connected H satisfies*

$$\text{sep}_{5/6}(H) \leq C_1 A(H) (\text{fsep}_{1/2}(H) + 1) \ln(2 + \text{fsep}_{1/2}(H)).$$

Proof. Apply rounding in each restriction, use structural heredity, and pass to the infimum by continuity and monotonicity. Then take the hereditary supremum. $\square$

Theorem 8.5 (Regularized Lean comparison). *There is a universal $C' > 0$ such that every H satisfies*

$$\text{fhw}(H) \leq C' L(H) (\text{adw}(H) + 1) \ln(2 + \text{adw}(H)).$$

For nonempty H , $L(H) \leq A(H) \leq 4\lambda(H)$. These inequalities, with separate estimates for $\text{adw}(H) \geq 2$ and $\text{adw}(H) < 2$, give the statement of Theorem 7.1.

Proof. First prove the connected bound and pass to primal components. Replace the component factor by $4\lambda(H)$. Bound $(a + 1) \ln(2 + a)$ by a constant times $a \log_2 a$ for $a \geq 2$, and by a constant for $a < 2$. The declaration `main_width_comparison` applies the proved rounding input to this deduction. $\square$

Corollary 8.6 (Auxiliary quantitative reverse comparison). *There is a universal constant $c > 0$ such that every finite hypergraph H satisfies*

$$\text{subw}(H) + 1 \geq \text{adw}(H) + 1 \geq c \frac{\text{fhw}(H)}{L(H) \ln(2 + \text{fhw}(H))}.$$

Proof. The parameterized Lean deduction uses the proved rounding theorem. The width hierarchy gives the first inequality and permits replacing $\ln(2 + \text{adw}(H))$ by $\ln(2 + \text{fhw}(H))$ in the upper comparison. Divide by the positive factors and take $c = C^{-1}$. $\square$